ORIE 4154: Pricing and Market Design **Fall 2026**

Lecture 15 — The Spiral-Down Effect

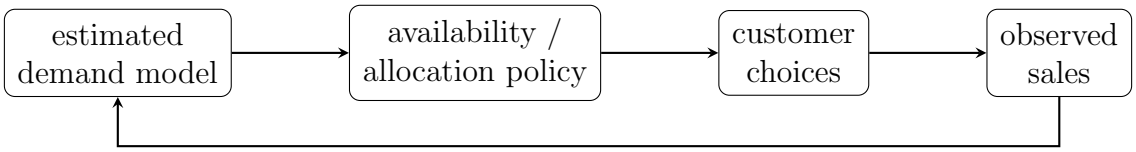
Lecturer: Sid Banerjee *Date: October 15, 2026 (planned)*

15.1 Overview

Planned for October 15. Lecture 14 on October 8 introduced substitution at the end of the Bayes Selector unit. This separate Spiral-Down draft develops that question; it is not a record of material already taught in Lecture 14.

Earlier lectures treated demand as an input to the allocation problem. Given a demand model, we used dynamic programming, fluid LPs, and compensated coupling to allocate scarce capacity.

In practice the model is estimated from historical sales. But historical sales were generated by yesterday's allocation policy. This creates a loop:



Our central question is:

> Can a revenue-management system make its own demand estimates wrong?

The answer is yes. A misspecified model of customer behavior, combined with feedback between estimation and optimization, can cause high-fare sales, protection levels, and revenue all to fall over time. This is the *spiral-down effect*.

The example is deliberately stark so that the general lesson is visible:

Sales are outcomes of both demand and the policy that shaped the choice set.

15.2 The hidden assumption behind Littlewood's rule

Consider one resource with capacity C and two fare classes:

$$p_1 > p_2.$$

Class 1 is the expensive fare and class 2 is the cheap fare. In the standard model, class-1 demand is a random variable D_1 with CDF F_1 and is independent of class-2 demand.

If x seats are already protected for class 1, the marginal value of protecting one more seat is

$$p_1 \mathbb{P}(D_1 > x) = p_1\big(1 - F_1(x)\big).$$

Littlewood's rule adds that seat when this exceeds the sure class-2 fare p_2. Thus one convenient integer-valued definition of the final protection level is

$$x^* = \min\left\{ x \in \{0, 1, \ldots, C\} : F_1(x) \geq 1 - \frac{p_2}{p_1} \right\}. \tag{15.1}$$

Thus the only demand input we need is the appropriate quantile of F_1.

But where does F_1 come from? A tempting procedure is:

1. observe expensive-fare sales on past departures;
2. treat those sales as samples of D_1;
3. form an empirical CDF $\widehat{F}_1$;
4. plug $\widehat{F}_1$ into (15.1).

This procedure assumes *perfect segmentation*: a class-1 customer wants only class 1 and a class-2 customer wants only class 2. Under perfect segmentation, closing class 2 does not change class-1 demand.

15.3 Customers choose among what is available

Real customers often have preference lists. Label the products 1 and 2, and use 0 for no purchase. Four simple customer types are:

$$102, \qquad 201, \qquad 210, \qquad 120.$$

The list 210, for example, means:

> buy class 2 if it is available; otherwise buy class 1; otherwise leave.

This is a price-conscious customer: the cheap product is preferred, but the expensive product is an acceptable substitute. A 120 customer has the opposite ranking and may trade down when class 1 is unavailable.

The important object is now not "demand for product 1" in isolation. It is a choice probability conditional on the offer set S:

$$q_i(S) = \mathbb{P}(\text{customer chooses } i \mid S \text{ is offered}).$$

In particular,

$$q_1(\{1\}) \quad \text{need not equal} \quad q_1(\{1,2\}).$$

Opening the cheap fare can divert sales away from the expensive fare.

Remark 15.1 (Censoring versus substitution). *If a product is unavailable, its sales are censored: sales can be below latent demand. Here there is a second issue. Availability changes which product customers choose. The observed product-level sales are therefore endogenous to the assortment or booking policy.*

15.4 A deterministic example

Suppose each departure has exactly $d < C$ customers, and every customer has preference list 210. A protection level x means:

- sell at most $C - x$ class-2 tickets;
- after class 2 closes, continue selling class 1.

Because $d < C$, everyone can be accommodated. The number of class-1 sales is

$$s(x) = \big[d - (C - x)\big]^+ = [d - C + x]^+, \tag{15.2}$$

and class-2 sales equal $d - s(x)$. Revenue is therefore

$$r(x) = p_1 s(x) + p_2\big(d - s(x)\big). \tag{15.3}$$

Since $p_1 > p_2$, revenue is increasing in $s(x)$ and hence in x. The true optimal policy is

$$\boxed{x^* = C.} \tag{15.4}$$

Close the cheap fare immediately; all d customers then buy the expensive fare.

This conclusion is not a recommendation about real airline pricing. It follows from the extreme assumption that every customer will buy class 1 when class 2 is closed. The extremity makes the feedback failure transparent.

15.5 The estimate–optimize loop

Now suppose the seller does not model choice. It interprets class-1 sales as independent samples of class-1 demand.

Let x_k be the protection level used on departure k. The seller observes

$$s_k = s(x_k) = [d - C + x_k]^+.$$

After k departures, it forms the empirical CDF

$$\widehat{G}_k(y) = \frac{1}{k} \sum_{j=1}^{k} \mathbf{1}\{s_j \le y\}. \tag{15.5}$$

Define the critical fractile

$$\gamma = 1 - \frac{p_2}{p_1}.$$

The next protection level is the empirical Littlewood quantile

$$x_{k+1} = \min \left\{ x \in \{0, 1, \dots, C\} : \widehat{G}_k(x) \ge \gamma \right\}. \tag{15.6}$$

Equations (15.2)–(15.6) describe a closed-loop system:

$$x_k \longrightarrow s_k \longrightarrow \widehat{G}_k \longrightarrow x_{k+1}.$$

The fatal error is the middle interpretation. The data point s_k is not autonomous class-1 demand. It is the number of customers forced to class 1 by the availability policy x_k.

15.6 Why protection spirals downward

Since $d < C$, whenever $x_k > 0$ we have

$$s_k = [d - C + x_k]^+ < x_k. \tag{15.7}$$

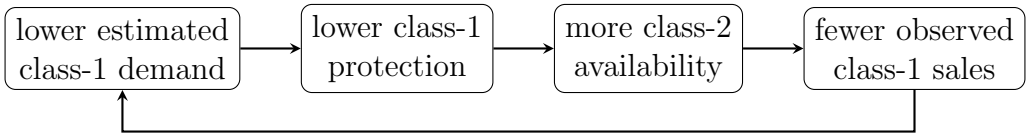
Thus the policy creates a new "demand observation" below its current protection level.

Proposition 1 (Spiral-down in the deterministic example). *Under the update rule (15.6), the sequence of protection levels is weakly decreasing. If the same update continues, it reaches 0 after finitely many departures.*

Proof (Proof sketch): Suppose x_k is the empirical γ-quantile of the previous observations. The next observation s_k lies strictly below x_k whenever $x_k > 0$. Adding an observation below the current quantile cannot increase the empirical quantile, so $x_{k+1} \leq x_k$.

If a positive level x remains fixed, then each additional observation equals $[d - C + x]^+ < x$. Eventually these observations constitute at least a γ fraction of the sample, forcing the empirical γ-quantile strictly below x. Since protection levels are nonnegative integers, repeated decreases must eventually reach zero. ■

The direction of the feedback is now visible:



More data can make a wrong model more precise without making it more correct. The observations become increasingly precise about the behavior induced by the current policy, while the model continues to attach the wrong meaning to them.

15.7 The numerical spiral

Use the values from Cooper, Homem-de-Mello, and Kleywegt:

$$C = 10, \qquad d = 8, \qquad p_1 = 500, \qquad p_2 = 200.$$

Then

$$\gamma = 1 - \frac{200}{500} = \frac{3}{5}.$$

Start at the true optimum $x_1 = 10$. The first departure closes class 2, so all eight customers buy class 1. The system records "class-1 demand equals 8" and lowers the next protection level to 8.

At $x = 8$, two cheap tickets are available. Two customers buy class 2 and the other six buy class 1. The system records class-1 demand of 6. The empirical 60th percentile eventually drops from 8 to 6. The process repeats.

departure k	protection x_k	class-1 sales	class-2 sales	revenue
1	10	8	0	4000
2	8	6	2	3400
3	8	6	2	3400
4	6	4	4	2800
⋮	⋮	⋮	⋮	⋮
8	6	4	4	2800
9	4	2	6	2200
⋮	⋮	⋮	⋮	⋮
20	4	2	6	2200
21	2	0	8	1600
⋮	⋮	⋮	⋮	⋮
50	2	0	8	1600
51	0	0	8	1600

The policy begins at the revenue-maximizing value

$$r(10) = 8(500) = 4000$$

and ends at the worst value

$$r(0) = 8(200) = 1600.$$

The firm "learns" its way from the best policy to the worst one.

15.8 What exactly went wrong?

The spiral requires two ingredients.

15.8.1 A misspecified behavioral model

The seller assumes separate, independent demands for the two fare classes. The actual customers choose from an availability set and substitute from class 2 to class 1 when class 2 is closed.

If the choice model were correct, a class-1 sale made after class 2 closes would be evidence of substitution, not evidence of an autonomous class-1 segment.

15.8.2 Feedback between estimation and optimization

The estimated model determines availability, availability determines sales, and those sales are reused to estimate the model. The data are therefore *policy-dependent*.

Either ingredient alone is less dangerous:

- a misspecified model used once may be biased but does not create an iterative collapse;
- feedback with a correctly specified choice model can support valid learning and control.

Together, they turn a local modeling error into a self-reinforcing system failure.

15.9 Connection to learning and experimentation

The issue resembles the bandit unit in Lectures 4–5: actions affect what data are observed. But the failures are different.

	Greedy bandit failure	Spiral-down failure
Unknown object	mean reward of each action	customer choice behavior
What actions change	which arm is sampled	which products are available and hence which sales occur
Core problem	insufficient exploration	wrong behavioral model plus endogenous data
Natural response	deliberate exploration / optimism	model choice sets; preserve variation; estimate choice conditional on availability

Randomization can help identify substitution. For example, occasionally varying which products are available lets us compare

$$q_1(\{1\}) \quad \text{with} \quad q_1(\{1,2\}).$$

But experimentation is not a substitute for specifying the relevant behavioral object. If the model insists that product-level demand is independent of the offer set, more carefully collected data will not cure the structural error.

15.10 The lesson beyond airlines

The same feedback loop appears whenever a policy controls which alternatives are visible or available:

- **Recommender systems:** rankings determine exposure, and clicks generated by that exposure are then used to update the ranking model.

- **Retail stockouts:** product availability determines substitution, so observed sales do not reveal what customers would have bought from a fully stocked shelf.
- **Platform search:** seller visibility affects transactions, and those transactions become inputs to future visibility and quality estimates.
- **Advertising allocation:** delivery policies determine which users see an ad, shaping the response data used to estimate future returns.

In every case,

$$\boxed{\text{model} \longrightarrow \text{policy} \longrightarrow \text{exposure or availability} \longrightarrow \text{observed data} \longrightarrow \text{model.}}$$

The application changes, but the diagnosis is the same: the data cannot be interpreted independently of the policy that generated them.

15.11 How choice models break the loop

A choice-based model begins with customer preferences and an offer set S. It predicts

$$q_i(S) = \mathbb{P}(i \text{ is chosen from } S).$$

Expected revenue from offering S is

$$r(S) = \sum_{i \in S} p_i q_i(S).$$

This representation makes substitution explicit: opening or closing one product may change the choice probabilities of every other product.

In the spiral-down example,

$$q_1(\{1\}) = 1, \qquad q_1(\{1,2\}) = 0,$$

for a type-210 customer. Once the system conditions on the offer set, the apparently contradictory sales data become perfectly coherent.

This leads to the subsequent choice-model unit:

1. build tractable random-utility and multinomial-logit models of substitution;
2. choose an assortment while accounting for how the offer set reshapes demand.

15.12 Takeaways

1. **Demand is not sales.** Sales combine preferences, availability, capacity, and the policy that generated the offer set.
2. **Product-level demand may not exist independently of the offer set.** Customers substitute.
3. **Optimization changes the data-generating process.** Re-estimating from policy-dependent data creates feedback.
4. **More data can reinforce a wrong model.** Statistical precision does not repair structural misspecification.
5. **Prices and availability also experiment.** Variation in offer sets can reveal substitution when analyzed with an appropriate choice model.

Scale can reduce sampling noise; it cannot remove policy-induced selection or structural misspecification.

Model customer choice first; then optimize using the demand that the policy itself creates.

Recommended reading

Core / original source.
Cooper, Homem-de-Mello, and Kleywegt, "Models of the Spiral-Down Effect in Revenue Management," *Operations Research* 54(5), 968–987, 2006. Read the introduction and motivating examples for the original phenomenon; the full stochastic analysis is optional.

Accessible companion.
Banerjee, *Intermission: The Spiral-Down Effect*, ORIE 4154 notes (online course note). This gives a shorter route through the protection-level example used here.

Bridge to choice models.
Talluri and van Ryzin, Sections 2.6.1–2.6.2, introduce buy-up factors and discrete-choice capacity models; Section 7.2.2.3 develops multinomial logit. Read these as preparation for modeling $q_i(S)$ in a later lecture, not as required proofs for the spiral-down result.